

Python Interview Coding Questions

1. Write a program to find the factorial of a number using a loop.
2. Check if a given string is a palindrome.
3. Find the largest of three numbers.
4. Reverse a string without using slicing.
5. Count vowels and consonants in a string.
6. Remove duplicates from a list.
7. Find the second largest number in a list.
8. Sort a list without using sort().
9. Count frequency of each element in a list.
10. Find common elements in two lists.
11. Write a recursive function to calculate factorial.
12. Fibonacci series using recursion.
13. Sum of digits of a number using recursion.
14. Check if a number is prime using a function.
15. Check for anagram strings.
16. Remove all characters except alphabets.
17. Find the first non-repeating character in a string.
18. Compress a string using character counts (Run-Length Encoding).
19. Write a list comprehension to create a list of squares of even numbers.
20. Invert a dictionary {key: value} to {value: key}.
21. Count characters in a string using dictionary comprehension.
22. Read a file and count the number of words, lines, and characters.
23. Write a program to copy contents from one file to another.
24. Create a class to represent a bank account with deposit and withdraw methods.
25. Demonstrate inheritance with a shape class (e.g., Circle inherits Shape).
26. Implement method overriding and use of super().
27. Handle division by zero error.
28. Custom exception class to validate user age.
29. Implement a stack and queue using a list.
30. Check balanced parentheses using a stack.
31. Implement a simple linked list or binary search.
32. Write a decorator to measure the execution time of a function.
33. Write a generator for Fibonacci series.
34. Implement a generator to yield even numbers up to N. Logical & Pattern Programs
35. Print Pascal's triangle.

36. Print diamond/star patterns.
37. Create a number pyramid.
38. Find the missing number in a list of 1 to N.
39. Rotate a matrix 90 degrees.
40. Implement binary search.
41. Merge two sorted lists.
42. Find pairs in a list that sum to a target.